THE UNIVERSITY OF SYDNEY
Account
Help
Dashboard
Courses
Groups
Calendar
Inbox
History
Studio
Student Portal
Support

ELEC1601 (ND) > Pages > Project Essentials
Immersive Reader

Semester 2 2026
Home
Ed Discussion
Modules
Assignments
Quizzes
Recorded Lectures
Zoom
People
Grades
Gradescope
Sparkplus
IgniteAI Search
Unit Outline (ELEC1601)
Unit Outline (ELEC9601)
Notebook

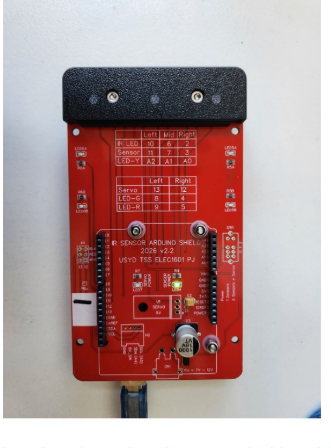
Project Essentials

▶ Overview

▶ Sensor shield

▼ The take-home board

To assist you with the essential task, we have created take-home boards as follows:



These boards emulate the sensor shield available on the robot.

With this board, you should be able to complete the "detection" part of each of the following scenarios at home.

Aspects to be aware of:

- The sensors are similar to those on the robot, but slightly different (they are horizontally instead of vertically aligned).
  - You may need to do some recalibration in the lab
  - You may need to do some code adjustment in the lab
- A pair of red and green LEDs has been provided on each side of the board (between the Arduino footprint and the front sensor block). You could use these to indicate whether you are commanding the servos to rotate forward (green) or backwards (red).
- A sheet of white paper behaves similar to the melamine we use in the maze
- The take-home board doesn't have wheels, but don't forget about them!!
  - Wheels extend approx. 2cm either side of the robot
  - You don't need to worry about left/right movements
- The gap between the walls in a corridor is 20cm

4. The 'Take-Home Robot Emulator'

Good news! You get to use our new take-home robot emulator!
While they don't have servos, a chassis, or wheels, they will allow you to develop robot code outside the classroom!
The difference?

- Wheels have been replaced with blinking LEDs (different frequencies for different speeds). Green for forward (LED5A and 6A), Red for backwards (LED5B and 6B).
- You will have to change 1 line of code to switch between the lab robot and the robot emulator.
- The IR LEDs and sensors are slightly better on the emulator.

The rest is pretty much the same.
Here's how to use it:

- Go to your lab demonstrator, tell them your team number, and sign off on receiving one.
  image.png
- Take it to wherever you meet your teammates.
  Caution: Don't lose it - we'll have to charge $300 for lost and severely damaged boards. This is how much it costs us to replace them.
- Download this compressed folder:
  The compressed folder contains 4 files: an example Arduino code scaffold *TakeHomeBoard_BasicFunctions.ino*, the library header file *ELEC1601_TakeHomeBoard.h*, the library implementation file *ELEC1601_TakeHomeBoard.cpp*, and the library configuration file *library.properties*.

  You don't need to adjust any of these files. However, **you do have to include it as a library in your Arduino IDE**:

  image.png

  Screenshot 2026-04-18 161734.png
  Screenshot 2026-04-18 161747.png
- **In your robot code, comment out the command *#include "Servo.h"* using 2 forward slashes and include the following command *#include "ELEC1601_TakeHomeBoard.h"*.**
- Use the USBA-to-USBB cable to connect the Take Home Robot Emulator to your computer. The cable supplies enough power for operating the board. If you do not have USBA ports on your device, you can use a dongle or a USBC-to-USBB cable.
- You're all set! You can upload your code and use a sheet of paper to simulate walls. Observe how your robot reacts and develop your code.
  Once you think your code is ready, save it, upload it to a lab robot in your next session, and test it on the maze. Make sure to comment out *#include "ELEC1601_TakeHomeBoard.h"* and include *#include "Servo.h"* instead when using the actual lab robot.

▼ The essential tasks - scenarios you must detect and escape from

For each of the following scenarios:

- Your robot must display on the three LEDs a binary code to indicate the scenario it has detected.
- Your robot must do nothing for at least 5 seconds to so that we can see this code being displayed on the LEDs
- After this, LEDs can display whatever you like
- Your robot must complete a sequence of servo operations to get out of that position.
- You can get marked off for each scenario individually, however, in order to be ready for extension, it is advised to have one piece of code that can detect and deal with each individual scenario

All of the scenarios can be tested in the maze at the start of this page

▼ 0: Unknown Scenario

What you must display:

- LED-R Right (A0): OFF
- LED-R Mid (A1): OFF
- LED-R Left (A2): OFF
- (After this, your LEDs can do what you want)

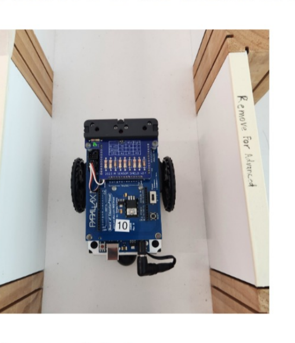
What your robot must do to:

- Not move at all.

Why this scenario is important:

- This is perhaps the most important scenario!
- If your robot doesn't know where it is, it is invaluable for it to stop
  - You can then photograph the position that your robot stopped in
  - You can then recreate it and decide what you want your robot to do

▼ 1: Robot is in the middle of a long corridor



What you must display:

- LED-R Right (A0): ON
- LED-R Mid (A1): OFF
- LED-R Left (A2): OFF
- (After this, your LEDs can do what you want)

What your robot must check:

- Both left and right sensors should identify a similar reading
  - You can choose how you measure this. Calibrating your sensors to measure distance is advised
- The forward sensor should see no wall, or a wall at least 10cm away.
- Once again, calibrating your sensors to measure distance is advised

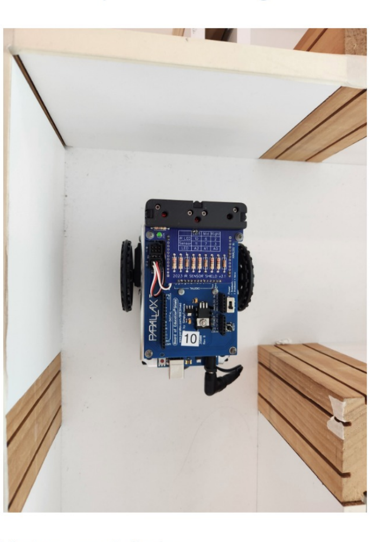
What your robot must do to:

- It must move 5cm forward
- It must remain approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- This is the main way your robot should move (along a corridor in a straight line)

▼ 2: Ideal position for Right Turn



What you must display:

- LED-R Right (A0): OFF
- LED-R Mid (A1): ON
- LED-R Left (A2): OFF
- (After this, your LEDs can do what you want)

What your robot must check:

- The right sensor should see no wall.
- The left sensor should see a wall
  - Ideally you are able to decide the wall is a specifc range away (at least x cm away, but less than y cm). This will help you not turn too early/late
- The forward sensor should see a wall
  - Ideally you are able to decide the wall is a specifc range away (at least x cm away, but less than y cm). This will help you not turn too early/late

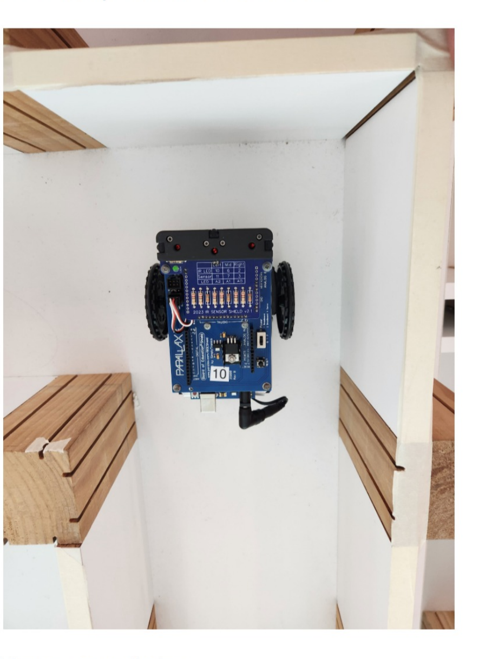
What your robot must do to:

- Robot must rotate 90 degrees
- It must then move forward.
- It should travel far enough that it re-enters the corridor and remains approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- This is how you will turn corridors

▼ 3: Ideal position for Left Turn



What you must display:

- LED-R Right (A0): ON
- LED-R Mid (A1): ON
- LED-R Left (A2): OFF
- (After this, your LEDs can do what you want)

What your robot must check:

- The left sensor should see no wall.
- The right sensor should see a wall
  - Ideally you are able to decide the wall is a specifc range away (at least x cm away, but less than y cm). This will help you not turn too early/late
- The forward sensor should see a wall
  - Ideally you are able to decide the wall is a specifc range away (at least x cm away, but less than y cm). This will help you not turn too early/late

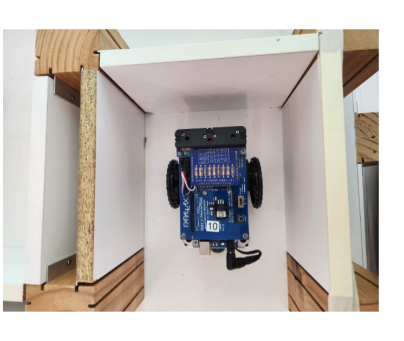
What your robot must do to:

- Robot must rotate 90 degrees
- It must then move forward.
- It should travel far enough that it re-enters the corridor and remains approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- This is how you will turn corridors

▼ 4: Dead End



What you must display:

- LED-R Right (A0): OF
- LED-R Mid (A1): OFF
- LED-R Left (A2): ON
- (After this, your LEDs can do what you want)

What your robot must check:

- The left, right and forward sensors should see all see a wall.
  - Ideally you are able to decide the wall is a specifc range away (at least x cm away, but less than y cm). This will help you verify it is indeed the end of the maze

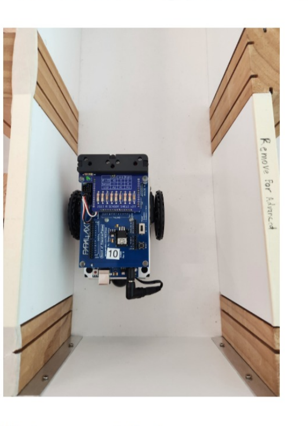
What your robot must do to:

- Robot must rotate 180 degrees
- It must then move forward at least 5cm.
- It should remains approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- At the halfway point to the extension maze, your robot must turn around

▼ 5: Bad Starting position - close to the left wall, but parallel to a wall



What you must display:

- LED-R Right (A0): ON
- LED-R Mid (A1): OFF
- LED-R Left (A2): ON
- (After this, your LEDs can do what you want)

What your robot must check:

- The left and right sensors should see a wall
  - You should be able to identify that the right sensor is significantly closer to the wall then the left sensor
  - You can choose how you measure this. Calibrating your sensors to measure distance is advised
- The forward sensor should see no wall, or a wall at least 10cm away.
  - Once again, calibrating your sensors to measure distance is advised

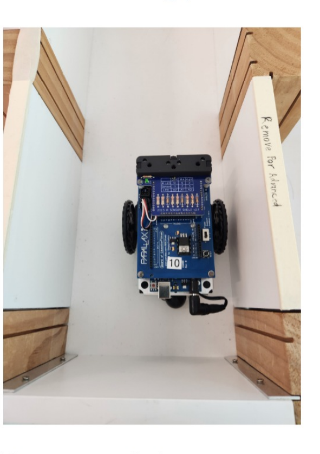
What your robot must do to:

- Robot must rotate clockwise slightly
- It must then move forward
- It should then rotate anticlockwise so it is once again parallel to the walls
- It should then move backwards so it is level with its starting position, but now more central
  - You may or may not find this step necessary in your final design. It can slow the robot down, but also reduce the likelihood of an accumulation of errors
- Its final position should be approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- Your robot must be robost to a bad initial starting position.

▼ 6: Bad Starting position - close to the right wall, but parallel to a wall



What you must display:

- LED-R Right (A0): OFF
- LED-R Mid (A1): ON
- LED-R Left (A2): ON
- (After this, your LEDs can do what you want

What your robot must check:

- The left and right sensors should see a wall
  - You should be able to identify that the left sensor is significantly closer to the wall then the right sensor
  - You can choose how you measure this. Calibrating your sensors to measure distance is advised
- The forward sensor should see no wall, or a wall at least 10cm away.
  - Once again, calibrating your sensors to measure distance is advised

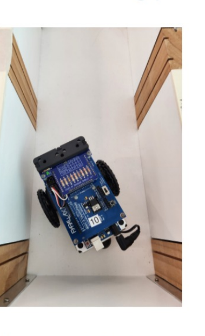
What your robot must do to:

- Robot must rotate anti-clockwise slightly
- It must then move forward
- It should then rotate clockwise so it is once again parallel to the walls
- It should then move backwards so it is level with its starting position, but now more central
  - You may or may not find this step necessary in your final design. It can slow the robot down, but also reduce the likelihood of an accumulation of errors
- Its final position should be approximately parallel to the walls
- The center of the robot must remain within 3cm of the center of the corridor
- Stop after completion

Why this scenario is important:

- Your robot must be robost to a bad initial starting position.

▼ 7: Bad Starting position - close to the left wall, approx 30° angle



What you must display:

- LED-R Right (A0): ON
- LED-R Mid (A1): ON
- LED-R Left (A2): ON
- (After this, your LEDs can do what you want)

What your robot must check:

- The left and forward sensors should see a wall relatively close
  - You can choose how you measure this. Calibrating your sensors to measure distance is advised
  - Be warned, the reading of the take-home board and robot will likely be different (sensors are vertical for the robot), so you might need to recalibrate
- The right sensor should see a wall far away
- You need to distinguish this from case 1!!
  - You must show your demonstrators that your robot can notice the difference

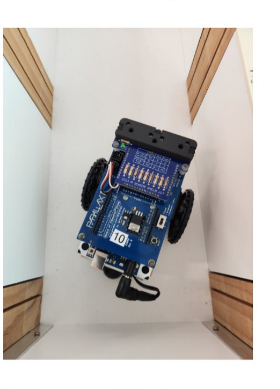
What your robot must do to:

- Robot must rotate clockwise slightly until it is parallel
- You could do nothing. Your robot might detect you are now in Scenario 5, or you create a custom solution.
- Stop after completion

Why this scenario is important:

- Your robot must be robost to a bad initial starting position.

▼ 8: Bad Starting position - close to the right wall, approx 30° angle



What you must display:

- LED-R Right (A0): FLASH (1 second delay)
- LED-R Mid (A1): OFF
- LED-R Left (A2): OFF
- (After this, your LEDs can do what you want)

What your robot must check:

- The right and forward sensors should see a wall relatively close
  - You can choose how you measure this. Calibrating your sensors to measure distance is advised
  - Be warned, the reading of the take-home board and robot will likely be different (sensors are vertical for the robot), so you might need to recalibrate
- The left sensor should see a wall far away
- You need to distinguish this from case 2!!
  - You must show your demonstrators that your robot can notice the difference

What your robot must do to:

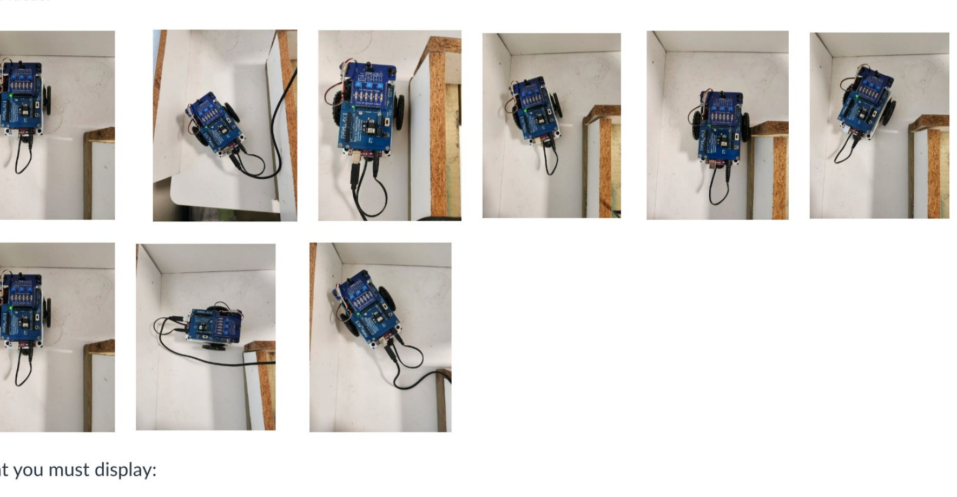
- Robot must rotate anticlockwise slightly until it is parallel
- You could do nothing. Your robot might detect you are now in Scenario 5, or you create a custom solution.
- Stop after completion

Why this scenario is important:

- Your robot must be robost to a bad initial starting position.

▼ 9/10: Your own scenarios

Some ideas:



What you must display:

- Scenario 9:
  - LED-R Right (A0): OFF
  - LED-R Mid (A1): FLASH (1 second delay)
  - LED-R Left (A2): OFF
  - (After this, your LEDs can do what you want)
- Scenario 9:
  - LED-R Right (A0): FLASH (1 second delay)
  - LED-R Mid (A1): FLASH (1 second delay)
  - LED-R Left (A2): OFF
  - (After this, your LEDs can do what you want)

What your robot must check:

- You decide

What your robot must do to:

- You decide
- Stop after completion

Why these scenarios are important:

- You need to learn and resolve your own issues

▼ Design Advice

It is not possible to take robots home -> You can only test your code in the labs. Therefore, the first thing you need to do is to separate work that can be done at home from work that must be done in the lab

▶ Work that can be done at home

▶ Work that must be done in the lab

Copyright © The University of Sydney. Unless otherwise indicated, 3rd party material has been reproduced and communicated to you by or on behalf of the University of Sydney in accordance with section 113P of the Copyright Act 1968 (Act). The material in this communication may be subject to copyright under the Act. Any further reproduction or communication of this material by you may be the subject of copyright protection under the Act. Do not remove this notice.
Live streamed classes in this unit may be recorded to enable students to review the content. If you have concerns about this, please visit our student guide and contact the unit coordinator.
Privacy Statement

◂ Previous
Next ▸